

The Magnetic Vector Potential

$\vec{\nabla} \times \vec{E} = 0$ allows us to write $\vec{E} = -\vec{\nabla} V$
in electrostatics.

V is defined upto a constant.

$\vec{\nabla} \cdot \vec{B} = 0$ allows us to write $\vec{B} = \vec{\nabla} \times \vec{A}$

If a vector with zero curl is added to ~~the~~ $\vec{A}$, that vector potential will give the same $\vec{B}$ -field. $\Rightarrow$ ~~the~~ Large freedom in choice of $\vec{A}$. ~~the~~ Often called "gauge" freedom.

$\vec{A} \rightarrow \vec{A} + \vec{\nabla} f$ is a "gauge" transformⁿ.

Notice $\vec{\nabla} \times \vec{A} = \vec{\nabla} \times (\vec{A} + \vec{\nabla} f)$ b/c $\vec{\nabla} \times \vec{\nabla} f = 0$

A common choice: impose $\boxed{\vec{\nabla} \cdot \vec{A} = 0}$
Coulomb condition / gauge

$\Rightarrow$ even with this condition, many choices of $\vec{A}$ will produce same $\vec{B}$ -field.

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$\vec{E}$ and $\vec{B}$ fields are "physical" (measurable).

V and $\vec{A}$ potentials are mathematical constructs.
(~~Not~~ Not measurable but very important.)

* With the Coulomb condition, $(\vec{\nabla} \cdot \vec{A} = 0)$, we get

$$\vec{\nabla} \times \vec{B} = \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} = -\nabla^2 \vec{A}$$

So Ampere's law becomes

$$\boxed{\nabla^2 \vec{A} = -\mu_0 \vec{J}}$$

Ampere's law
with Coulomb
condition.

* Calculating electric and magnetic potentials.

Electric potentials are determined by

$$\boxed{\nabla^2 V = -\frac{\rho}{\epsilon_0}}$$

Poisson's Equation.

Obtained by combining
 $\vec{E} = -\vec{\nabla} V$ and $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

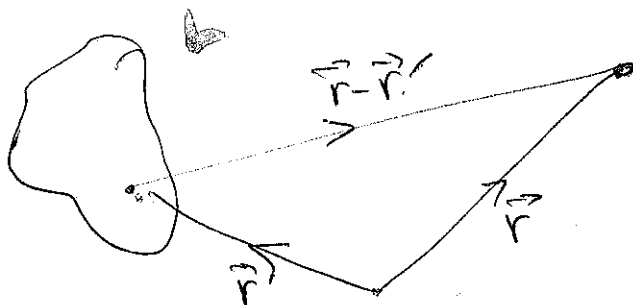
$$\text{and } \vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f$$

In free space ($\rho=0$), $\boxed{\nabla^2 V = 0}$ Laplace's Equation.

* Solving Poisson's equation [obtaining $V(\vec{r})$ for a given $\rho(\vec{r})$] is a rich topic — many methods exist. General solution:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i(\vec{r}_i)}{|\vec{r} - \vec{r}_i|} \quad \text{for point charges}$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3\vec{r}'$$



Many approximations and tricks exist.

[MULTIPOLE EXPANSION, METHOD OF IMAGES, ...]

* Ampere's law with Coulomb condition:

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}$$



Each Cartesian component is a Poisson's equation:

$$\nabla^2 A_x = -\mu_0 J_x, \quad \nabla^2 A_y = -\mu_0 J_y, \quad \nabla^2 A_z = -\mu_0 J_z$$

$$\Rightarrow \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3\vec{r}' \quad \text{Same methods can be used.}$$